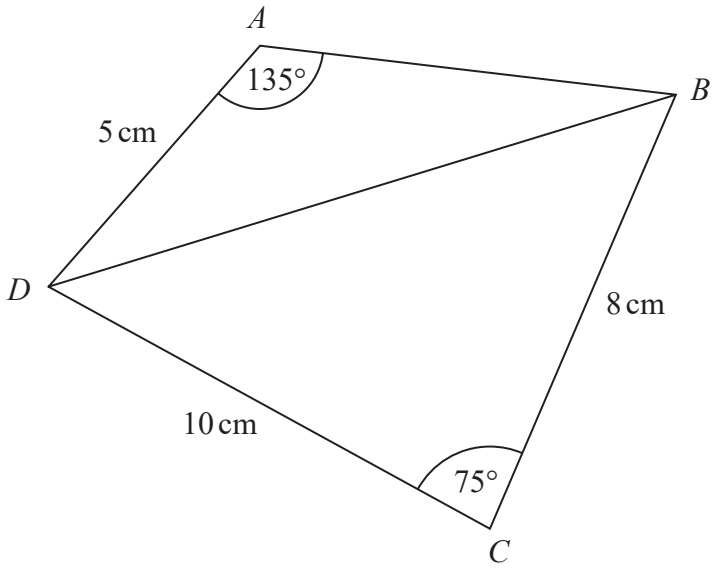
17 *ABCD* is a quadrilateral.



Calculate the size of angle *ABD*.
Give your answer correct to 1 decimal place.

In BCD: $BD^2 = 10^2 + 8^2 - 2 \times 10 \times 8 \times \cos 75°$

1 mark
Cosine rule: use the included 75° angle.

$BD = \sqrt{122.58895...} = 11.071989...$ cm

1 mark
Take the square root; retain full precision.

In ABD, use the sine rule:

1 mark
Match each angle with its opposite side.

$$\frac{\sin(\text{angle ABD})}{5} = \frac{\sin 135°}{11.071989...}$$

Angle ABD $= \sin^{-1}(5 \sin 135° \div 11.071989...)$

$= 18.621949...° \approx 18.6°$

1 mark
ABD must be acute: the other angle in ABD is already 135°.

18.6 °

(Total for Question 17 is 4 marks)

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